Attempts on all eight research questions

Proofs, partial results, a literature update, and the remaining gaps
31 October 2026

No unrestricted open problem is claimed solved. This report works through every public question on the supplied site. It gives self-contained proofs for restricted cases and limitations, and identifies a recent coloring preprint that postdates the site's review. Several arguments are standard; proving them here does not establish novelty.

The strongest directly proved result concerns the candidate graph-network question: a local, fixed-order two-layer model has an explicit root-n sampling rate without an analytic activation. A separate construction shows why unrestricted continuous activations cannot share one convergence rate across general architectures.

Question	Outcome of this attempt
1. Testing → distance estimation	General conversion remains unresolved; an edgeless-property example proves that a quadratic sample loss can be necessary.
2. Sharp k-colorability samples	A new preprint reports O(k log k / ε); the factor-k gap remains. A multipartite subclass is analyzed sharply below.
3. Polynomial-time general GI	Unresolved. An explicit pair defeats spectra and ordinary color refinement as complete invariants.
4. GI fixed-parameter in degree	Unresolved. A complete algorithm is given for few branching vertices, a different parameter.
5. Cospectral graphon approximation	Unresolved in general; proved positive for complete bipartite graphons with isolates.
6. Triangle-removal quantitative gap	No new unrestricted bound. Exact tripartite behavior and the progression-free construction are derived.
7. Polynomial induced-C4 removal	Unresolved in general; a polynomial $\varepsilon^5/4$ guarantee is proved for triangle-free input graphs.
8. Higher-order network transfer	Explicit local and polynomial-model rates; an arbitrary-slow example for general continuous activation.

Conventions: simple undirected graphs; one unordered edge edit counts once; edit distance is divided by n^2. Triangles and induced C4 copies are counted as unordered vertex sets. Sample complexity counts vertices, not inspected pairs or computation time. If a requested sample exceeds the graph order, read the whole graph.

1. Testing versus distance estimation

Question. Can the sample complexity of estimating distance to every testable dense-graph property be bounded by the tester complexity evaluated at a polynomially smaller accuracy parameter?

The general doubly exponential conversion and the sharper target remain separated in the sources checked [S1]. The calculation below establishes a necessary loss in a particularly simple property; it does not disprove the permitted polynomial rescaling.

Proposition 1. A quadratic loss is sometimes necessary

Let E be the property of having no edges. Its optimal testing sample complexity is $\Theta(1/\varepsilon)$, while estimating its normalized edit distance at arbitrary thresholds needs $\Theta(1/\varepsilon^2)$ sampled vertices in the worst case.

$$q_E(\varepsilon) = \Theta(\varepsilon^{-1}), \quad s_E(\varepsilon) = \Theta(\varepsilon^{-2}).$$

Testing upper bound. Draw R independent ordered vertex pairs, allowing repeated vertices. If an ε-far graph has $m \geq \varepsilon n^2$ edges, each pair is an edge with probability $2m/n^2 \geq 2\varepsilon$. Reject on the first edge. With $R = \lceil \log(3)/(2\varepsilon) \rceil$, failure is at most $\exp(-2\varepsilon R) \leq 1/3$. At most 2R vertices are used. A canonical sample can simulate these draws: expose a uniformly ordered set of distinct vertices as new labels are first encountered, reproducing repeats internally.

Testing lower bound. Give a proportion a of the vertices all possible neighbors, and leave the remaining vertices independent. Its edge density m/n^2 tends to $d(a) = a - a^2/2$. For $a = 2\varepsilon$ and large n, it is ε-far. A sample of s vertices misses the core with probability tending to $(1-a)^s$, and then is identical to an edgeless sample. Distinguishing the two distributions with constant advantage requires $s = \Omega(1/\varepsilon)$.

Estimation upper bound. For the same independent pairs let Z be the adjacency indicator, including zero on repeated endpoints. Then $E[Z]/2 = m/n^2$ exactly. Hoeffding's inequality gives an additive $O(\varepsilon)$ estimate from $O(\varepsilon^{-2})$ pairs, hence from $O(\varepsilon^{-2})$ vertices.

Estimation lower bound. Compare the core constructions with $a_0 = 1/4$ and $a_1 = 1/4 + 2\varepsilon$, for $0 < \varepsilon \leq 1/16$. Their limiting distances differ by more than ε. The entire induced sample is a deterministic function of s core-membership bits. The relative entropy of their bit distributions is $s \cdot \mathrm{KL}(\mathrm{Ber}(a_0) \,||\, \mathrm{Ber}(a_1)) = O(s\varepsilon^2)$. Data processing and Pinsker's inequality imply total variation $O(\varepsilon\sqrt{s})$; constant success therefore requires $s = \Omega(\varepsilon^{-2})$. Uniform samples without replacement converge to this experiment as n grows, which supplies finite-graph lower bounds.

What this resolves, and what it leaves

A conversion preserving the same accuracy up to a constant factor is impossible for optimal sample complexities. The edgeless property does obey the desired rescaling, since $q_E(\varepsilon^2)$ has the right order. A general proof or a property separating all polynomial rescalings is still missing.

2. Sharp samples for k-colorability

Question. Is the optimal one-sided canonical sample size $\Theta(\log(k)/\varepsilon)$, with uniform dependence on both parameters?

Literature update after the website review

Kushnir–Shapira's published September 2026 bounds are $\Omega(\log(k)/\varepsilon)$ and $O((k/\varepsilon)\log(1/\varepsilon))$ [S2]. Xu's preprint, submitted 22 October and revised 26 October 2026, reports $O(k \log(k)/\varepsilon)$ [S3]. It uses a palette process with a block encoding. This is a recent preprint, not an independently established result of this report. Its claimed consequence is optimal $\Theta(1/\varepsilon)$ dependence for each fixed k; a factor k remains in the joint dependence.

Published: $\Omega(\log(k)/\varepsilon) \le S_k(\varepsilon) \le O((k/\varepsilon)\log(1/\varepsilon))$.
New preprint reports: $S_k(\varepsilon) \le O(k \log(k)/\varepsilon)$.

Proposition 2. A sharp complete-multipartite subclass

Restrict inputs to complete (k+1)-partite graphs, with no additional isolated vertices. For $0 < \varepsilon \le 1/(16k^2)$, the worst-case number of sampled vertices needed to detect non-k-colorability is, over arbitrarily large blowups,

$$\Theta\left(1/(k\varepsilon) + \log(k)/\sqrt{\varepsilon} \right).$$

This includes constants uniform in $k \ge 2$. Finite-order rounding is handled by rational approximations with slack, or by taking sufficiently large blowups.

Distance formula. Let the positive part proportions, in increasing order, be $p_1,\dots,p_{k+1}$, with sum one. There is an optimal k-coloring that is constant on each original part: conditional on all other assignments, the monochromatic-edge cost is linear in the allocation of an independent part among the colors, so moving the whole part to a cheapest color cannot increase the cost. Repeating this for each part proves the assertion. Some two parts must share a color, and merging the two smallest is optimal. Therefore

$$d_{k\text{-col}}(G) = p_1 p_2.$$

Upper bound. A sample is non-k-colorable exactly when it meets every part. If $p_1 p_2 \ge \varepsilon$, then $p_2 \ge \sqrt{\varepsilon}$. Also $p_2 \le 1/k$, so $p_1 \ge k\varepsilon$. The probability of missing a part of mass p is at most $\exp(-sp)$, also for sampling without replacement. Thus

$$\Pr[\text{miss some part}] \le \exp(-sk\varepsilon) + k \exp(-s\sqrt{\varepsilon}).$$

Taking s at least $\max\{\log(6)/(k\varepsilon), \log(6k)/\sqrt{\varepsilon}\}$ makes this at most 1/3.

Matching lower bounds

For the first term, take $p_1 = 2k\varepsilon$ and all other parts of size $(1-2k\varepsilon)/k$. These inputs are ε-far in the stated range, and seeing the smallest part with constant probability needs $\Omega(1/(k\varepsilon))$ samples.

For the second term, take k parts of mass $\sqrt{\varepsilon}$ and one remaining part of mass $1-k\sqrt{\varepsilon}$. The distance is ε. Collecting all k rare parts requires $\Omega(\log(k)/\sqrt{\varepsilon})$ samples. For example, the number of missing rare parts has mean $k(1-\sqrt{\varepsilon})^s$ and variance at most its mean, because distinct missing-part indicators have nonpositive covariance. For s below a sufficiently small constant times $\log(k)/\sqrt{\varepsilon}$, the probability of collecting all parts is bounded away from 2/3. The maximum of these two lower bounds is comparable to their sum.

2. Where the general coloring argument stops

The multipartite calculation removes a logarithm in ε in a concrete class, but its examples are too structured to realize the general Ω(log(k)/ε) lower bound. The September paper obtains that lower bound from sparse random graphs followed by blowups [S2]. A coupon-collector analysis of complete multipartite graphs cannot settle the full problem.

An attempted way to remove the factor k

The new palette argument charges degree thresholds to deleted vertex–color labels. There are initially kn such labels. A tempting modification is to charge to palette entropy, whose initial size is about n log(k), hoping to replace k by log(k).

The missing implication is reconstruction. A potential based on reciprocal palette sizes controls weighted color-availability degrees. It does not directly control the number of monochromatic edges in the coloring reconstructed later. For instance, k available colors at every neighbor can divide a weighted degree by k, while a deterministic assignment can still put all those neighbors in the same color. Replacing the kn budget alone therefore does not prove the conjectured bound.

Precise unresolved claim

Neither the recent reported upper bound nor Proposition 2 proves O(log(k)/ε) for arbitrary graphs with an absolute constant. A reconstruction argument converting a smaller potential into an unweighted edge-error guarantee would be a substantive next step, and is not supplied here.

How to interpret the recent preprint

The revised source was read, including its reconstruction, palette invariants, deletion-stability, and final counting steps. This report records its stated theorem and separates it from the published benchmark. The website’s displayed upper bound should be supplemented by this preprint, with its status visible. No change to the website was made.

3. Polynomial-time general graph isomorphism

Question. Is there a deterministic $n^{O(1)}$-time isomorphism test for all finite simple graphs? The quasipolynomial algorithm remains the general benchmark in the primary sources checked [S4, S5]. No new general algorithm is proved here.

A concrete failure of a natural invariant approach

Consider the 4×4 rook graph R, whose vertices are pairs in $\mathbb{Z}_4^2$ and whose edges join different vertices in the same row or column. Consider also the Cayley graph S on $\mathbb{Z}_4^2$ with connection set $\{(1,0),(3,0),(0,1),(0,3),(1,1),(3,3)\}$. This is the Shrikhande graph.

Both are 6-regular on 16 vertices. For every distinct pair of vertices, adjacent or not, the number of common neighbors is two. Consequently both adjacency matrices satisfy

$$A^2 = 4I + 2J, \quad \operatorname{Spec}(A) = \{6^{(1)}, 2^{(6)}, (-2)^{(9)}\}.$$

The all-ones vector has eigenvalue 6. On its orthogonal complement J vanishes, giving eigenvalues ±2. The trace and dimension determine the indicated multiplicities.

Nevertheless the graphs are not isomorphic. R has a clique of order four in each of its four rows and four columns. S has no clique of order four: the six neighbors of any vertex induce a 6-cycle, which has no triangle, whereas a four-clique would give such a triangle. Thus their four-clique counts are eight and zero, respectively.

Ordinary color refinement also gives the same single color at every round, since both graphs are 6-regular. Combining this refinement with the entire adjacency spectrum still identifies these nonisomorphic graphs as indistinguishable.

Result of the attempt

This is an exact counterexample to using those invariants as a complete polynomial-time solution. It is not a lower bound against every polynomial-time algorithm. Adding selected motif counts distinguishes this pair, but no proof is given that a polynomial-size list suffices for all graphs.

Computational certificate

The verification program constructs both 16×16 matrices, checks $A^2 = 4I + 2J$ using integer arithmetic, and counts every four-vertex clique exactly. The eigenvalue calculation is numerical corroboration; the displayed spectrum follows exactly from the matrix identity.

4. Isomorphism fixed-parameter in maximum degree

Question. Can the time be $f(d)\cdot n^{C}$ for maximum degree d, with C independent of d? The $n^{O((\log d)^{c})}$ bound [S6] still has a degree-dependent exponent; the 2026 survey explicitly leaves this parameterization open [S5].

Proposition 3. Few branching vertices admit an elementary FPT algorithm

Let B be the vertices of degree at least three, and let $t = |B|$. Graph isomorphism is decidable in $t!\cdot n^{O(1)}$ time. This parameter t is different from maximum degree.

Algorithm. If the two branch sets have different sizes, reject. Otherwise try every bijection π between them and check that it preserves the induced graph on B. Remove B. Every remaining component is a path, an isolated vertex, or a cycle.

A cycle component has no neighbors in B, because all its vertices already have degree two. Record its length. For a path on at least two vertices, internal vertices have no neighbors in B, and each endpoint has at most one. Record the path length and the unordered pair of endpoint attachments, using a null label for an unattached end. For an isolated vertex, record its set of zero, one, or two neighbors in B. Transform these attachment labels using π and compare the multisets of all records.

Correctness. Any isomorphism maps B to B and induces one of the enumerated bijections. It then preserves every recorded component. Conversely, matching records give path isomorphisms, with reversal when needed, cycle isomorphisms, and singleton matches. Together with π these form an isomorphism of the full graphs.

Time. Each branch bijection permits a polynomial-time comparison of the records. The n factor needed to list and sort records has an absolute exponent. This gives the claimed FPT time.

Why this does not answer the listed question

When $d \le 2$, $t = 0$ and the algorithm reduces to comparing path and cycle length counts. Such counts can be collected in linear time. But a cubic graph can have $t = n$, so the branching-vertex algorithm does not yield $f(d)\cdot n^{C}$.

Likewise, one cannot algebraically absorb an unbounded exponent $a(d)$ in $n^{a(d)}$ into $f(d)$: for any fixed C and any fixed d with $a(d) > C$, the ratio $n^{a(d)-C}$ is unbounded as n increases. This rules out a common misreading of the existing bound; it does not rule out a better algorithm.

5. Finite cospectral approximation of graphons

Question. Does there exist an equal-edge-density cospectral graphon pair that is not the cut-distance limit of equal-order, exactly cospectral finite graph pairs? The cited paper poses this explicitly [S7, S8]. The general existence question remains unresolved in this report.

The known obstruction and why it disappears

Finite cospectral graphs of common order have equal edge counts, because $\operatorname{tr}(A^2) = 2|E|$. Edge density is continuous in cut distance. Hence any simultaneous limits must have equal edge density. Also, equal graphon cycle densities fix the nonzero operator eigenvalues and therefore $\int W^2$; they do not by themselves fix $\int W$. Once the latter is assumed equal, this particular obstruction vanishes [S8].

Proposition 4. Complete bipartite graphons are always approximable

Let $B_{a,b}$ be one on the two rectangles between disjoint sets of measures a and b, and zero elsewhere, with $a+b \le 1$. The complement of those sets consists of isolates. The only nonzero eigenvalues are $\pm\sqrt{(ab)}$, and its edge density is $2ab$.

Thus $B_{a,b}$ and $B_{c,d}$ are cospectral precisely when $ab = cd > 0$; they automatically have equal edge density. Every such pair admits simultaneous finite cospectral approximation.

Proof. First shrink all four parameters by a common factor tending to one, so that both support sums are strictly less than one. Approximate the shrunken a,b,c by positive rationals a',b',c' and set $d' = a'b'/c'$. Then d' is rational and approaches the shrunken d. The strict support inequalities remain valid for sufficiently close approximations.

Choose a common denominator N, enlarge it by an arbitrary integer factor if necessary, and form

$$G = K_{Na',Nb'} \text{ plus } N(1-a'-b') \text{ isolates},$$
$$H = K_{Nc',Nd'} \text{ plus } N(1-c'-d') \text{ isolates}.$$

Both graphs have order N, and their two nonzero adjacency eigenvalues are $\pm N\sqrt{(a'b')} = \pm N\sqrt{(c'd')}$. Their zero multiplicities also agree. Their step graphons converge even in L^1 to the target graphons, hence in cut distance. Choosing N tending to infinity completes the construction. If $ab = 0$, both graphons are zero almost everywhere and the claim is trivial.

An exact nonisomorphic example

$$G_r = K_{2r,2r} + 4r \text{ isolates},$$
$$H_r = K_{4r,r} + 3r \text{ isolates}.$$

Their common order is 8r; their spectra are $\{2r, -2r, 0^{(8r-2)}\}$; their edge densities are 1/8. Their graphons are exactly $B_{1/4,1/4}$ and $B_{1/2,1/8}$. The degree-zero masses differ, so the limiting graphons are not weakly isomorphic. This verifies that nonisomorphic equal-density cospectral limits can be exactly approximable; it supplies no resistant pair.

Remaining obstruction to a general proof

Independent random approximations only match limiting spectral moments, not the entire finite characteristic polynomial. Turning that asymptotic agreement into exact finite cospectrality is the unproved step. The rational product construction works because the bipartite family has only one product constraint.

6. Quantitative triangle removal

Question. Improve the general asymptotic gap between a quasipolynomial obstruction and a tower-type guarantee. Fox's guarantee and Hunter's construction remain separated in the checked primary sources [S9, S10]. No unrestricted improvement is established below.

$$\exp(\Omega(\log^2(1/\varepsilon))) \le 1/\delta_{\triangle}(\varepsilon) \le \mathrm{tower}(O(\log(1/\varepsilon))).$$

Proposition 5. Exact behavior for a complete tripartite core

Let G consist of a complete tripartite graph with part proportions a,b,c and any remaining isolated vertices. Its normalized triangle-removal distance and triangle density are

$$\tau_{\triangle}(G)/n^2 = \min\{ab,ac,bc\}, \quad T(G)/n^3 = abc.$$

Proof of the distance. Write the integer part sizes as $A \le B \le C$. Deleting all A–B edges destroys every triangle, so $\tau \le AB$. There are AB edge-disjoint triangles: index vertices $i \in \{0,\dots,A-1\}$, $j \in \{0,\dots,B-1\}$, and use the C-part vertex $(i+j) \bmod C$. A fixed i sees distinct C-vertices as j varies; a fixed j sees distinct C-vertices as i varies. All three edge types are therefore used at most once. At least AB deletions are necessary.

Consequently, if this graph is ε-far, the three products are at least ε. Multiplying them gives $(abc)^2 \ge \varepsilon^3$, so

$$T(G) \ge \varepsilon^{3/2} n^3.$$

The exponent is exact within this class: take $a = b = c = \sqrt{\varepsilon}$ and isolate the remainder, for $\varepsilon \le 1/9$. Rational approximations give the same asymptotic statement for every small ε.

A self-contained form of the standard progression-free obstruction

Let $S \subseteq \{1,\dots,M\}$ contain no nonconstant three-term arithmetic progression, and let $s = |S|$. Make vertex parts $X = \{1,\dots,M\}$, $Y = \{1,\dots,2M\}$, $Z = \{1,\dots,3M\}$. For every $x \in X$ and $a \in S$, add the three edges of the triangle $(x, x+a, x+2a)$, with coordinates in X,Y,Z respectively.

Any triangle has differences $a = y-x$ and $b = z-y$ in S, while $c = (z-x)/2$ is in S. The relation $a+b = 2c$ forces $a=b=c$. Thus the graph has exactly Ms triangles, and they are edge-disjoint. Its order is 6M, giving exactly

$$\tau_{\triangle}/n^2 = s/(36M), \quad T/n^3 = s/(216M^2).$$

A progression-free set with $s/M \ge \exp(-C\sqrt{\log M})$ yields the familiar $\exp(\Omega(\log^2(1/\varepsilon)))$ obstruction along a sequence of ε values. Hunter's stronger constants use a more refined construction [S10]. The reduction here reproduces the standard scale and does not improve it.

Why the elementary packing route stops

In an arbitrary graph, a large edge-deletion distance can force many edge-disjoint triangles while still giving only order εn^2 triangles. The required removal conclusion is on the n^3 scale. The progression-free examples show why that missing factor cannot be recovered from disjointness alone.

7. Polynomial induced-C4 removal

Question. Does ε-farness from induced-C4-freeness imply at least $a\varepsilon^b n^4$ induced copies for absolute $a,b > 0$? The polynomial conjecture remains explicit in the checked notes; the general established guarantee is exponential in a polynomial of $1/\varepsilon$ [S11, S12].

Proposition 6. A polynomial guarantee for triangle-free input graphs

Suppose G is triangle-free and requires at least εn^2 arbitrary edge additions/deletions to become induced-C4-free, with $0 < \varepsilon \le 1$. If Q counts its induced C4 vertex sets, then

$$Q \ge (\varepsilon^5/4)\, n^4.$$

Step 1: distance supplies two lower bounds. Deleting every edge makes the graph induced-C4-free, hence $m = |E| \ge \varepsilon n^2$. Every ordinary four-cycle in a triangle-free graph is induced. Greedily deleting an edge from each remaining four-cycle cannot create a new four-cycle, so at most the initial Q deletions suffice. Therefore $Q \ge \varepsilon n^2$ as well.

Step 2: count common neighbors. For $u<v$ let c_{uv} be their number of common neighbors, put $N = n(n-1)/2$, and set

$$S = \sum_{u<v} c_{uv} = \sum_w \operatorname{binom}(\deg(w),2) \ge 2m^2/n - m.$$

For $n \ge \varepsilon^{-2}$, the right side is at least $\varepsilon^2 n^3$. Indeed the function $2m^2/n - m$ is increasing for $m \ge \varepsilon n^2$ in this regime, and $2\varepsilon^2 n^3 - \varepsilon n^2 \ge \varepsilon^2 n^3$. Thus $S/N \ge 2\varepsilon^2 n \ge 2$.

Each four-cycle is counted once for each of its two opposite pairs, so $Q = (1/2)\sum \operatorname{binom}(c_{uv},2)$. Cauchy–Schwarz gives

$$\begin{aligned} Q &= (1/4)\Big(\sum c_{uv}^{\,2} - S\Big) \\ &\ge (1/4)(S^2/N - S) \ge S^2/(8N) \ge (\varepsilon^4/4)n^4. \end{aligned}$$

Here $S/N \ge 2$ gives the middle inequality. Since $\varepsilon \le 1$, this is at least $(\varepsilon^5/4)n^4$.

Step 3: small graphs. If $n < \varepsilon^{-2}$, Step 1 gives $Q \ge \varepsilon n^2 > \varepsilon^5 n^4$, which is stronger than the claimed bound. This covers every n and completes the proof.

The exact gap to unrestricted input graphs

The argument uses triangle-freeness twice: ordinary four-cycles are induced, and their count is monotone under edge deletions. In a general graph, a four-cycle can have diagonals; deleting edges can create induced copies. A complete graph has many ordinary four-cycles and no induced C4 at all. The displayed counting identity for ordinary cycles therefore cannot be used as a bound on induced copies in the unrestricted question.

8. Quantitative graph-network transfer

Question. Give explicit rates for fixed higher-order invariant networks on independent graph-signals sampled from the same bounded graphon-signal. The cited paper proves qualitative transfer and a logarithmic quantitative bound for a local two-layer analytic class [S13]. We first analyze that local architecture directly.

Proposition 7. A root-n bound for every bounded local kernel

Fix k. Let h be any measurable real function of the adjacency indicators and signals in an ordered k-tuple, with $|h| \le M$. For a sampled simple loopless graph-signal on n vertices, set

$$N_n = n^{-k} \sum_{(i_1,\dots,i_k)\in[n]^k} h(A \text{ on the tuple; } f_{i_1},\dots,f_{i_k}).$$

Tuples with repeated vertices use the actual diagonal $A_{ii} = 0$ and repeated signal values. Let μ be the expectation of h on k distinct sampled vertices. For $n \ge k$,

$$E|N_n - \mu| \le M[k/\sqrt{n} + k(k-1)/n].$$

Consequently, for independently sampled graphs of orders $n,m \ge k$ from the same graphon-signal,

$$E|N_n - N_m| \le Mk(n^{-1/2} + m^{-1/2}) + Mk(k-1)(n^{-1} + m^{-1}).$$

Proof. Let U_n be the average over injective ordered tuples. Its expectation is exactly μ. Kernels supported on disjoint vertex sets are independent: their latent positions and all their internal Bernoulli edge variables are disjoint. The covariance of any two bounded kernels has absolute value at most M^2. Two uniformly chosen injective k-tuples overlap with probability at most k^2/n, by a union bound on their coordinates. Hence $\mathrm{Var}(U_n) \le M^2k^2/n$ and $E|U_n - \mu| \le Mk/\sqrt{n}$.

The proportion of noninjective tuples is $1-(n)_k/n^k \le \mathrm{binom}(k,2)/n$. Replacing their contribution by the injective average changes the statistic by at most twice M times this proportion. Thus $|N_n - U_n| \le Mk(k-1)/n$ deterministically. The triangle inequality proves both claims. No smoothness or analyticity is used.

Application to the paper's equation (182)

$$h = \rho\Big(\sum_{u<v} a_{uv}A_{uv} + \sum_u b_u f_u + c\Big).$$

When $|f| \le r$ and ρ is continuous, it is bounded on the reachable interval

$$I = \Big[c + \sum \min(a_{uv},0) - r\sum|b_u|,\; c + \sum \max(a_{uv},0) + r\sum|b_u|\Big].$$

Take $M = \max_{x\in I}|\rho(x)|$. Proposition 7 applies directly, giving the root-n bound for arbitrary fixed weights and continuous activation. Linear combinations of local channels satisfy the sum of their bounds, weighted by the absolute readout coefficients.

8. The limit and polynomial deeper networks

The sampling limit need not equal the network on W

For the local model, the target μ integrates the conditional expectation of h over independent Bernoulli edges with probabilities $W(x_u,x_v)$. It is not generally obtained by replacing every A in h by W. For example, averaging A_{ij}^2 equals averaging A_{ij} on a simple graph, so its limit is $\int W$, whereas applying the same square formula to W gives $\int W^2$. Proposition 7 concerns size transfer between sampled finite graphs, as requested.

The root-n order is sharp over the stated class. Take a network that ignores the edges and averages a signal equal to +1 or −1 on two equal-measure sets. Its expected absolute sample mean is asymptotic to $\sqrt{(2/\pi n)}$. Independent samples of equal order likewise have an absolute difference of order $n^{-1/2}$.

Proposition 8. An explicit bound for finite motif polynomials

Suppose the finite-graph output is $P = \sum_{j=1}^{J} c_j\, t(H_j,d_j;G,f)$, where H_j is a loopless multigraph on v_j vertices, d_j assigns nonnegative integer signal exponents, and D_j is their sum. A factor for each repeated edge uses that same adjacency entry. Put

$$A = \sum |c_j|\, r^{D_j}\, v_j,$$
$$B = \sum |c_j|\, r^{D_j}\, v_j(v_j-1).$$

For n,m at least the largest motif order,

$$E|P(G_n,f_n) - P(G_m,f_m)| \le A(n^{-1/2}+m^{-1/2}) + B(n^{-1}+m^{-1}).$$

Proof. A motif summand is a local kernel of absolute value at most r^{D_j}. Apply Proposition 7 to each term and sum using $|c_j|$. On distinct vertices, parallel adjacency factors collapse because $A_{uv}^{\ell} = A_{uv}$. Thus its deterministic sampling limit is the same expression with parallel edges removed from each motif.

Polynomial-activation IWNs expand into finite linear combinations of these signal-weighted motif densities; this representation is explained in [S13, Appendix G.3]. The displayed constants expose the coefficient, signal, and motif-order dependence. They apply at arbitrary fixed depth once that expansion is supplied.

A quantitative reduction for continuous activations

If a polynomial expansion P uniformly approximates N within η on all admissible finite graph-signals, the preceding estimate gives

$$E|N(G_n,f_n) - N(G_m,f_m)| \le 2\eta + A_\eta(n^{-1/2}+m^{-1/2}) + B_\eta(n^{-1}+m^{-1}).$$

This becomes computable when the approximation error and expanded coefficients are certified. For Lipschitz activations, if layer ℓ has activation constant L_ℓ, linear sup-norm bound K_ℓ, and a polynomial activation approximation error δ_ℓ on a validated interval containing both preactivations, errors propagate as $e_\ell \le L_\ell K_\ell e_{\ell-1} + \delta_\ell$. Include the readout norm. The motif counts and coefficients may grow quickly; this reduction does not establish a tight architecture-level rate.

8. Continuity permits arbitrarily slow transfer

Proposition 9. There is no common rate shape for all continuous activations

Let r_n be any positive nonincreasing sequence tending to zero. There is a single fixed continuous bounded activation σ and a fixed IWN, ignoring edges, whose independent-sample transfer errors are not $O(r_n+r_m)$. The constant in such an O-bound cannot rescue it, even if allowed to depend on that network.

Construction. Use $W = 0$ and the signal $f = +1$ on the first half of the unit interval and -1 on the second. A normalized invariant linear layer computes the sample mean S_n; apply σ and an identity linear readout. Integration to a scalar is one of the allowed linear equivariant operators, so this is an IWN. Choose increasing integers n_j such that $r_{n_j} \le 4^{-j}$, and put $a_j = (2n_j)^{-1/2}$.

Define a continuous nondecreasing g on $[0,1]$ with $g(0)=0$ and $g(a_j)=2^{-j}$, using linear interpolation between successive points and a bounded extension above a_1. Set $\sigma(t)=g(|t|)$, extended continuously and boundedly outside $[-1,1]$. This activation is fixed once the sequence is chosen.

A uniform fluctuation probability. For the Rademacher sample mean, $E[S_n^2]=1/n$ and $E[S_n^4]=(3n^2-2n)/n^4 \le 3/n^2$. Applying Paley–Zygmund to S_n^2 yields

$$\Pr(|S_n| \ge 1/\sqrt{(2n)}) \ge 1/12.$$

Therefore $E[\sigma(S_{n_j})] \ge 2^{-j}/12$. By the law of large numbers, continuity at zero, and boundedness, $E[\sigma(S_m)] \to 0$. For each j choose $m_j \ge n_j$ so large that this expectation is at most $2^{-j}/24$. Then

$$E|\sigma(S_{n_j})-\sigma(S_{m_j})| \ge 2^{-j}/24,$$
$$r_{n_j} + r_{m_j} \le 2\cdot 4^{-j}.$$

The ratio is at least $2^j/48$ and diverges. This proves the proposition. The same conclusion applies to any rate tending to zero by choosing the subsequences to make both rate values small; monotonicity is convenient, not essential.

Consequences for the candidate question

There is an explicit and sharp root-n answer for the fixed-order local class, and an explicit coefficient-dependent answer for polynomial deeper networks. A common rate over all continuous activations is impossible. General tight rates can still depend on activation moduli, tensor orders, depth, and weights; obtaining such a sharp architecture-level theorem remains unfinished. The counterexample does not rule out a root-n theorem for appropriately normalized Lipschitz architectures.

Verification, interpretation, and remaining work

What was checked computationally

The accompanying program uses only Python and NumPy, with no network access. It verifies the finite examples and boundary conventions; the general statements rely on the written proofs.

• Rook and Shrikhande matrices: exact integer matrix identities, equal spectra, and exact four-clique counts 8 versus 0.
• Multipartite distance: exhaustive unrestricted vertex colorings for 24 small part-size instances, plus 60 checks of the sample upper bound.
• Eight exact complete-bipartite cospectral pairs, of orders 8 through 64.
• 125 tripartite edge-disjoint triangle packings and a rational equality example.
• All 64, 1,024, and 32,768 labeled graphs on 4, 5, and 6 vertices: induced-C4 edit distance computed by breadth-first search in the full edge-edit cube; the triangle-free subclass bound is checked on every eligible graph.
• Exact binomial sums and fourth moments for the signal sampling examples.

The induced-C4 exhaustive calculation allows both additions and deletions. It is not a deletion-only proxy. Computed eigenvalues are supplemental; exact cospectrality of the bipartite family follows from its displayed characteristic polynomial, and that of the 16-vertex example follows from the integer matrix identity.

What remains unsolved

No universal tester-to-estimator rescaling is proved. The coloring conjecture still needs its uniform dependence on k closed. Neither unrestricted polynomial GI nor FPT in degree is established. No resistant equal-density cospectral graphon pair is constructed, and no universal positive approximation theorem is proved. The triangle-removal gap is unchanged. Polynomial induced-C4 removal remains unproved beyond the restricted input class above. Tight explicit rates for every fixed deep continuous-activation IWN are not derived.

Evidence and novelty

A literature search is not a proof that no newer solution exists. The stated status refers to the primary sources and targeted later-work searches checked on 31 October 2026. The new coloring bound is explicitly attributed to a recent preprint. The propositions here are mathematical derivations, not claims of publication novelty, peer review, or formal verification. Several are elementary or standard special cases.

Reproducibility bundle

The archive contains this report in searchable offline HTML, the verification program, its JSON results, and a source manifest. The program checks finite instances and does not attempt a brute-force proof of an asymptotic conjecture. The public website was read but not modified.

Primary sources

All links were checked or read in this session on 31 October 2026. Locations identify the relevant statements; online versions may subsequently change.

[S1] Gishboliner and Shapira, Polynomial Property Testing (2025)
Section 5, Theorem 5.3 and Problem 5.4.
Open primary source

[S2] Kushnir and Shapira, Towards Tight Bounds for Testing k-Colorability (2026)
Theorems 1.1-1.2 and Conjecture 1.3.
Open primary source

[S3] Xu, An Improved Bound for Testing k-Colorability (2026), version 2
Theorem 1.1; Sections 4-7; revised 26 October 2026.
Open primary source

[S4] Babai, Graph Isomorphism in Quasipolynomial Time
General graph isomorphism upper bound.
Open primary source

[S5] Neuen, Parameterized complexity of graph isomorphism testing (2026)
Introduction and Section 4.1, after Corollary 4.7.
Open primary source

[S6] Grohe, Neuen and Schweitzer, A Faster Isomorphism Test for Graphs of Small Degree
Maximum-degree running time.
Open primary source

[S7] Hladký, Iľkovič, León and Shu, On Cospectral Graphons (2026)
Journal Problem 7; preprint Problem 4.4.
Open primary source

[S8] On cospectral graphons, preprint
Section 4, unequal-density obstruction and Problem 4.4.
Open primary source

[S9] Fox, A new proof of the graph removal lemma (2011)
Triangle-removal upper bound.
Open primary source

[S10] Hunter, An improved construction for the triangle removal lemma
Introduction and construction.
Open primary source

[S11] Gishboliner, Removal Lemmas: Summer School 2025
Section 3, Conjecture 3.6.
Open primary source

[S12] Gishboliner and Shapira, Efficient Removal without Efficient Regularity
Induced C4 exponential removal guarantee.
Open primary source

[S13] Herbst and Jegelka, Higher-Order Graphon Neural Networks: Approximation and Cut Distance (ICLR 2025)
Definition 4.3; Theorems 5.2, 5.4; Appendices G.3-G.4, equation (182).
Open primary source